Method of analysis for enzyme activity assay for enterokinase:

1. Reagent/ Chemicals used:

Table 1: Reagents and chemicals			
S. No:	**Name**	**Make**	**Catalogue Number**
1	Trypsinogen	Sigma	T1143
2	HCl	Merck	100317
3	$CaCl_2$	SRL	31944
4	Succinic acid	Sigma	398055
5	BAEE.HCL	Sigma	B4500
6	Sodium phosphate monobasic anhydrous	Sigma	39638
7	Sodium hydroxide	SRL	68151

2. Equipments/ Instruments used:

Table 2: Equipments/ Instruments			
Serial no.	**Name**	**Make**	**Equipment/Instrument ID**
1.	UV Spectrophotometer	Jenway Genova Plus	BG-SPM-011
2.	Adjustable mono-channel and multi-channel pipettes	Eppendorf	BG-PIP-090, BG-PIP-085, BG-PIP-073
3.	Waterbath	Ascension Innovations	BG-WBH-004

3. Reagent Preparations:

- **Reagent A – 40 mM Succinate Buffer, pH 5.6 at 25 °C** Dissolve 472.36 mg of succinic acid in 100 ml of deionized water. Adjust the pH to 5.6 at 25 °C using 1 M NaOH.
- **Reagent B – 1 mM Hydrochloric Acid with 5 mM Calcium Chloride Solution** Mix 1 ml of 0.1 N hydrochloric acid and 73.505 mg of calcium chloride dihydrate in 100 ml of deionized water.
- **Reagent C – 0.1% (w/v) Trypsinogen Solution** Immediately before use, prepare reagent C by dissolving trypsinogen at a concentration of 10 mg/ml in cold reagent B.
- **Reagent D – Enterokinase Enzyme Solution** Immediately before use, prepare a solution containing 2.5 units/ml of enterokinase in cold deionized water. Dilute samples to the required concentrations using buffer.

- **Reagent E – 67 mM Sodium Phosphate Buffer, pH 7.6 at 25 °C** Dissolve 8.0387 g of sodium phosphate monobasic, anhydrous in 1 L of deionized water. Adjust the pH to 7.6 at 25 °C using 1 M NaOH.
- **Reagent F – 0.248 mM Na-Benzoyl-L-Arginine Ethyl Ester Solution (BAEE)** Prepare freshly by dissolving 8.502 mg of Na-benzoyl-L-arginine ethyl ester hydrochloride in 100 ml of reagent E.
- **Reagent G – 40 mM Hydrochloric Acid with 5 mM Calcium Chloride Solution** Mix 40 ml of 1 M hydrochloric acid and 735.05 mg of calcium chloride dihydrate in 1 L of deionized water.

4. Protocol:

Step 1:

- Pipette 1.80 mL of Reagent A and 0.50 mL of Reagent C into a clean test tube
- Mix thoroughly by inversion and equilibrate at 25 °C
- Add 0.10 mL of Reagent D to the mixture
- For each condition:
  - **Standard**: Add 0.10 mL of prepared standard solution
  - **Test**: Add 0.10 mL of prepared sample dilution
  - **Blank**: Add 0.10 mL of deionized water
- Mix all tubes by inversion and incubate at 25 °C for 15 minutes
- Stop the reaction by adding 3.00 mL of Reagent G to each tube

Step 2:

- Pipette 3.00 mL of Reagent F and 0.20 mL of the test mixture into a quartz cuvette
- Mix immediately by inversion
- Record UV absorbance at 253 nm continuously for 5 minutes
- Maintain the temperature of the mixture at 25 °C throughout the measurement
- **Calculations for enzyme activity**

$$Units/mL = \frac{(\Delta A_{253nm}/min\ test - \Delta A_{253nm}/\min blank)\ (5.4)(df)}{(0.001)(0.20)(P.A.)(0.024)(15)(0.1)}$$

5.4 - Volume (in milliliters) of Step 1
df - Dilution factor
0.001 - The change in A253nm/minute per unit of Trypsin as per the Unit Definition
0.20 - Volume (in milliliter) from Step 1 used in Step 2
P.A. - Potential activity of Trypsinogen1
0.024 - mg trypsin/nanomole trypsin
15 - Time (in minutes) for Step 1 as per the Unit Definition
0.1 - Volume (in milliliter) of enterokinase used